Trend Following, Leveraged Re-Entry, and Volatility Decay

S&P 500 total return. One signal only: the 200-day moving average (the daily equivalent of Faber's 10-month rule). Tested as far back as the data allows.

Data. Total return throughout. Monthly S&P 500 total return back to 1901 (Shiller) for the Faber replication; a daily total-return series from 1928 for everything else — real `^SP500TR` from 1988, and before that `^GSPC` price plus the Shiller dividend yield (this reconstruction tracks the real series with 0.50%/yr error and 0.9996 correlation over 1988–2026). Cash and the financing of leverage use **real T-bill rates** throughout: `^IRX` (13-week T-bill) from 1960, and the Ken French / Ibbotson **1-month T-bill** (monthly) before that, back to 1926 — so the risk-free covers the entire 1928+ daily sample (it averaged ~0% in the 1930s–40s, ~1–3% in the 1950s, and ~3.2% over the full sample). The trend signal is lagged one day, so nothing uses information we could not have had.

How the ratios are computed (all from *daily* returns, then annualized):

- **CAGR** — geometric: the constant yearly rate that turns the start wealth into the end wealth. "Grew $1 to" is the literal end value of $1.
- **Volatility** — annualized standard deviation of daily returns (× √252).
- **Sharpe** = annualized **mean** daily excess return over the T-bill ÷ volatility. It uses the *arithmetic mean*, **not** the CAGR. For high-volatility strategies the arithmetic mean sits far above the CAGR (by ≈ ½·vol², the variance drag), so Sharpe will **not** equal (CAGR − rf)/vol — e.g. the 4×-above rule's arithmetic mean is ~35% vs a 20.7% CAGR, giving Sharpe ≈ 0.56, not (20.7% − 3%)/52% ≈ 0.34. This is the standard definition and is correct; the volatility column lets you check it directly from the arithmetic mean.
- **Sortino** — same as Sharpe but the denominator counts only downside deviation (shortfalls below zero, averaged over *all* days).
- **Calmar** = CAGR ÷ |max drawdown|.
- **Information ratio (IR vs S&P)** = annualized mean of (strategy − S&P buy-&-hold) daily return ÷ its standard deviation (the tracking error). It is undefined for buy-and-hold itself (it *is* the benchmark).

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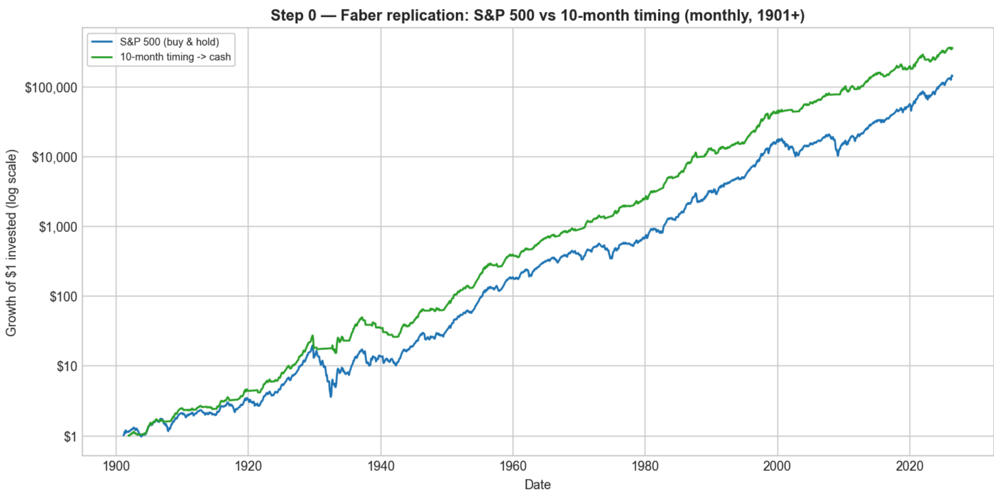
1. Buy & hold vs the Faber moving-average rule

The rule: hold the S&P 500 while it is **above** its moving average; move to **cash** while it is **below**. Faber uses the 10-month SMA on monthly data; we use that, then its daily twin (the 200-day SMA).

Monthly, 1901–2026 (Faber's exact setup):

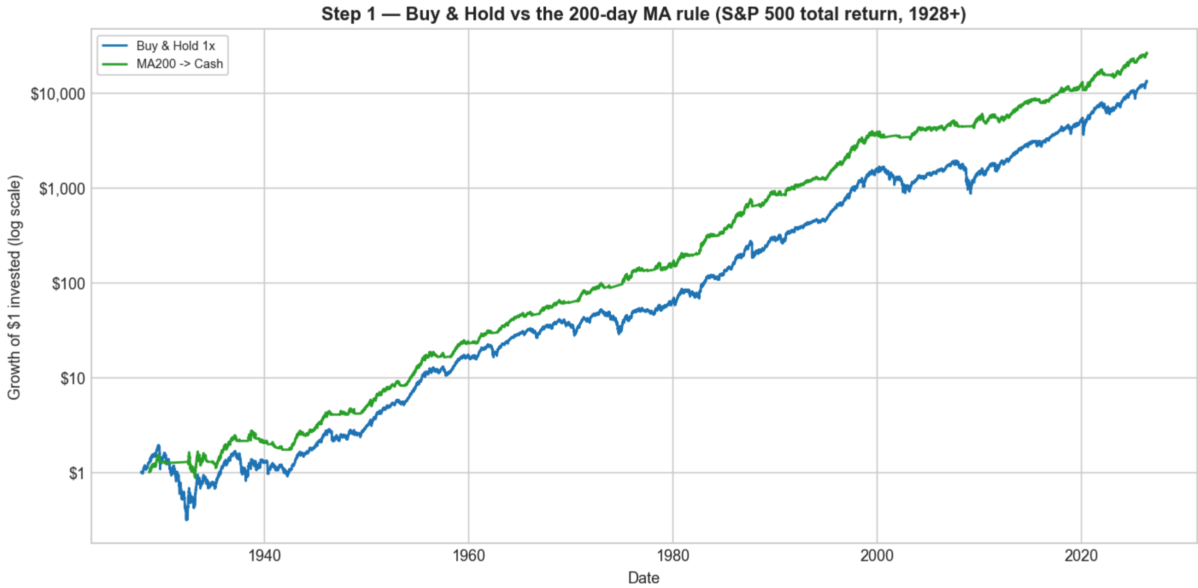
	S&P 500 buy & hold	10-month timing → cash
CAGR	9.95%	10.89%
Volatility	15.4%	10.8%
Sharpe	0.50	**0.74**
Max drawdown	−81.8%	**−47.5%**

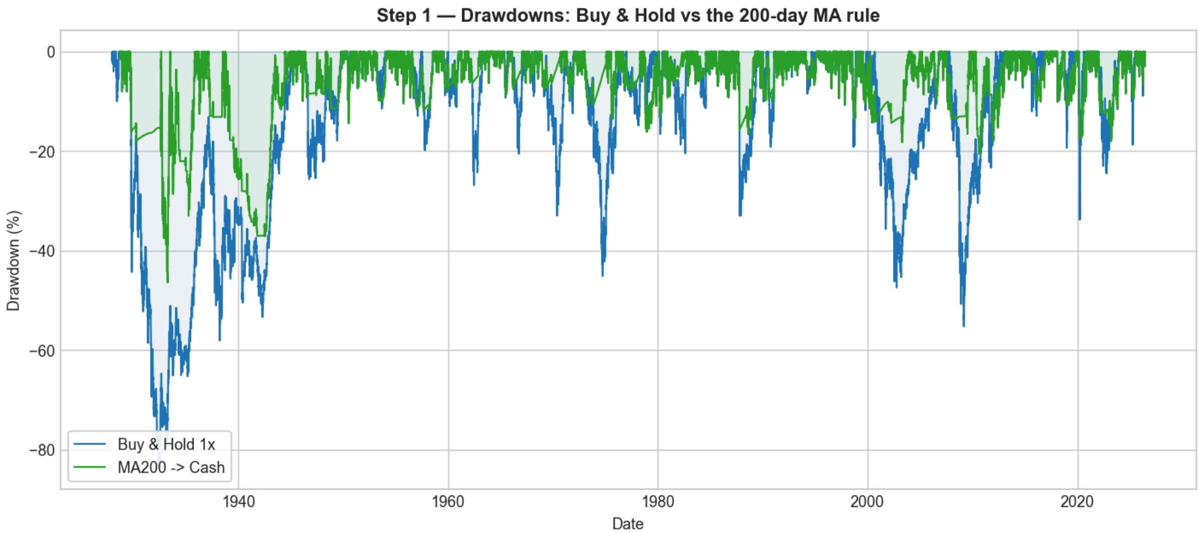
(Faber's published drawdowns are −83.66% → −42.24%; we reproduce the same qualitative halving — small differences come from extending the sample to 2026 and using real, near-zero 1930s–40s cash rates.)



Daily, 200-day SMA, 1928–2026:

	Buy & Hold 1×	MA200 → Cash
CAGR	10.14%	11.01%
Volatility	18.9%	12.6%
Sharpe	0.43	**0.63**
Sortino	0.62	**0.90**
Max drawdown	–83.9%	**–46.4%**
Calmar	0.12	**0.24**





The moving-average rule keeps essentially all of the return while cutting volatility by a third and halving the worst drawdown, so its Sharpe, Sortino and Calmar are all much higher. The 200-day MA adds clear risk-adjusted value.

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2. Leverage returns

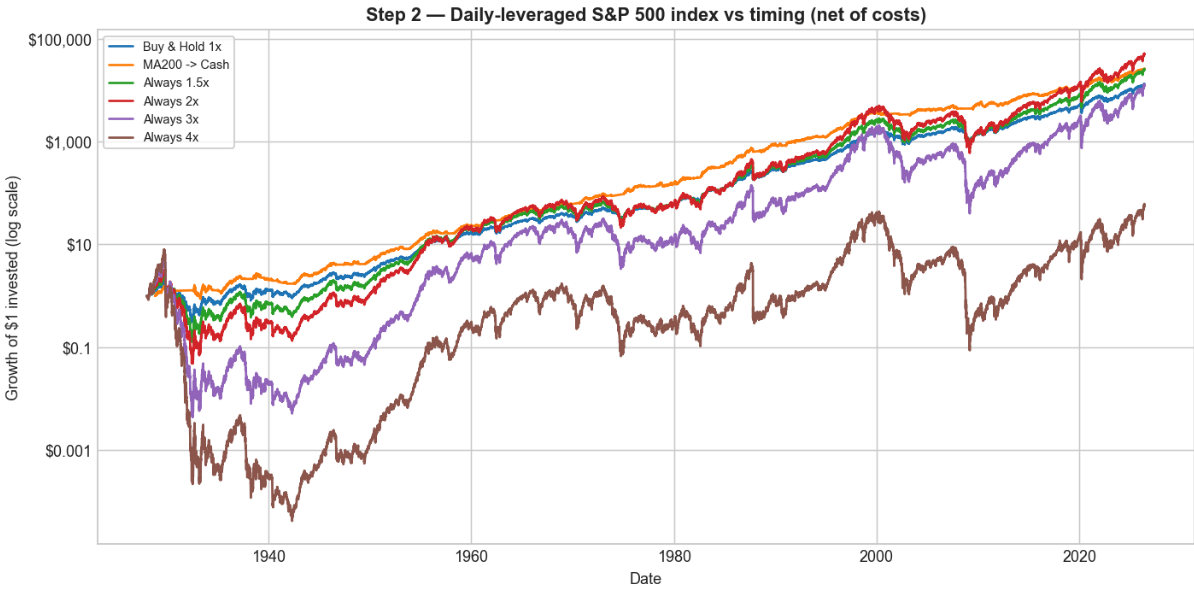
A daily **L× leveraged** return is simply that day's S&P 500 total return multiplied by L, and then compounded day by day:

```
leveraged_return[t] = L × sp500_return[t]        (before fees / financing)
```

For example, the 2× series is just the daily S&P total return × 2, compounded. Borrowed money costs the financing rate, and leveraged funds charge a fee (~0.9%/ yr); both are included below.

Holding **constant** daily leverage on the index, net of costs:

	CAGR	Grew $1 to
1× (buy & hold)	10.14%	$13,021
Always 1.5×	10.93%	$25,037
Always 2×	11.59%	$49,549
Always 3×	10.09%	$12,094
Always 4×	**4.20%**	**$54**



Constant leverage helps a little up to ~2×, then **runs out of road**: constant 3× only matches buy & hold and constant 4× collapses to $54 (CAGR 4.2%) — across a full century (including the 1929, 1987 and 2008 crashes) the volatility drag and deep drawdowns overwhelm the extra exposure. Section 6 shows that *switching* the leverage with the trend fixes this.

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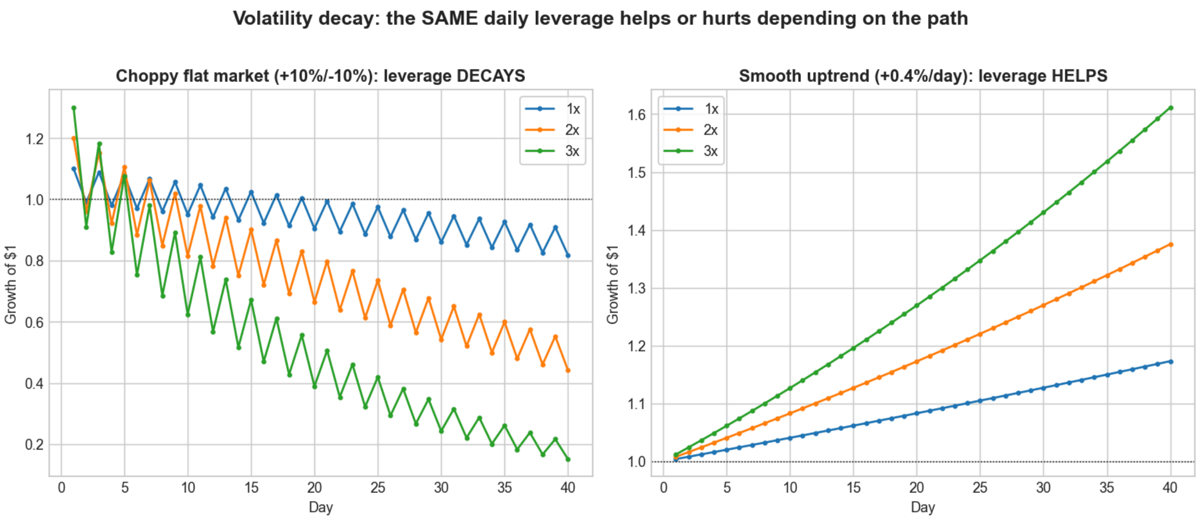
3. Volatility decay, and buying leverage at the lows

Volatility decay. A +10% day followed by a −10% day:

	1×	2×	3×
Two-day return	−1.0%	−4.0%	−9.0%

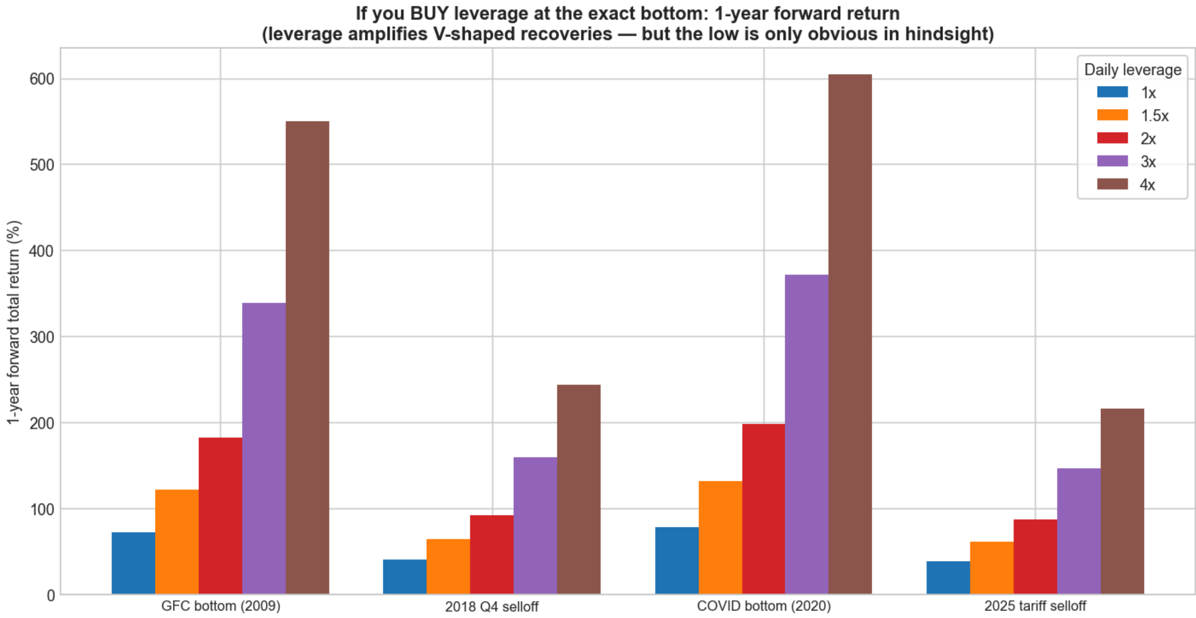
The market round-trips to roughly flat, but leverage loses — and the loss grows with the *square* of leverage. The annual penalty is ≈ $\frac{1}{2} \cdot L^2 \cdot \sigma^2$: at 20% volatility, ~2%/yr for 1×, **8%/yr for 2×, 18%/yr for 3×**.



But decay only bites in choppy/falling markets. In a one-directional rally — like the rebound off a market bottom —

leverage amplifies the gain. Forward **1-year** total return if you had bought at the exact low:

Bottom	1×	1.5×	2×	3×	4×
GFC (2009-03-09)	+72%	+122%	+182%	+339%	**+550%**
2018 Q4 (2018-12-24)	+40%	+64%	+92%	+159%	**+244%**
COVID (2020-03-23)	+78%	+132%	+198%	+372%	**+605%**
2025 tariff selloff (2025-04-08)	+39%	+61%	+87%	+146%	**+216%**



Leverage can work strongly in your favour **if you time it** — buying into the recovery off a low. (The low, of course, is only obvious in hindsight.)

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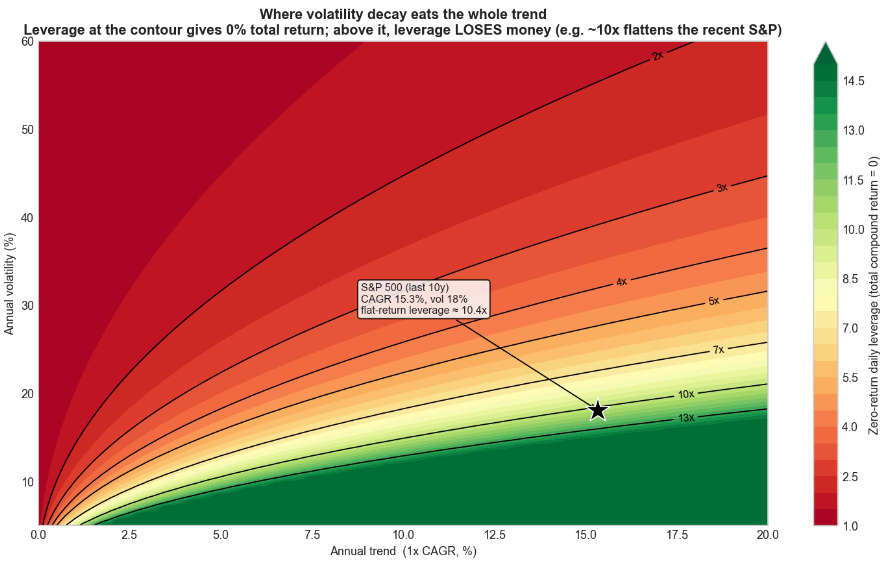
4. The leverage at which the total return goes flat

For a given trend (the 1× CAGR g) and volatility σ, the compound return of L× leverage is $L \cdot \mu - \frac{1}{2} \cdot L^2 \cdot \sigma^2$. Setting it to zero gives the leverage at which volatility decay exactly cancels the trend, so the **total return is flat (0%)**:

$$L_{zero} = 2 \cdot g / \sigma^2 + 1$$

Below it, leverage still grows; above it, leverage **loses money**. The map below plots L_{zero} for every combination of trend and volatility. For the S&P over the **last 10 years** (CAGR 15.3%, volatility 18.1%) the flat point is **≈ 10×** — so a hypothetical "10× S&P" would have gone essentially nowhere despite a strong decade, while higher still would have bled toward zero.



(For reference, the leverage that merely *ties* 1× buy & hold is lower — ≈ 3.1× for the S&P — and the growth-maximising level is ≈ 2×; see `charts/F3_breakeven_leverage_map.png` and `charts/F3_optimal_leverage_curve.png`.)

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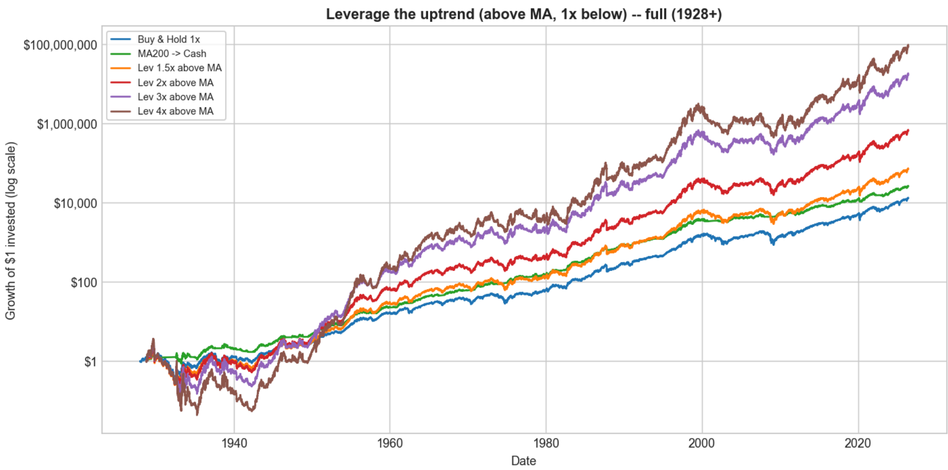
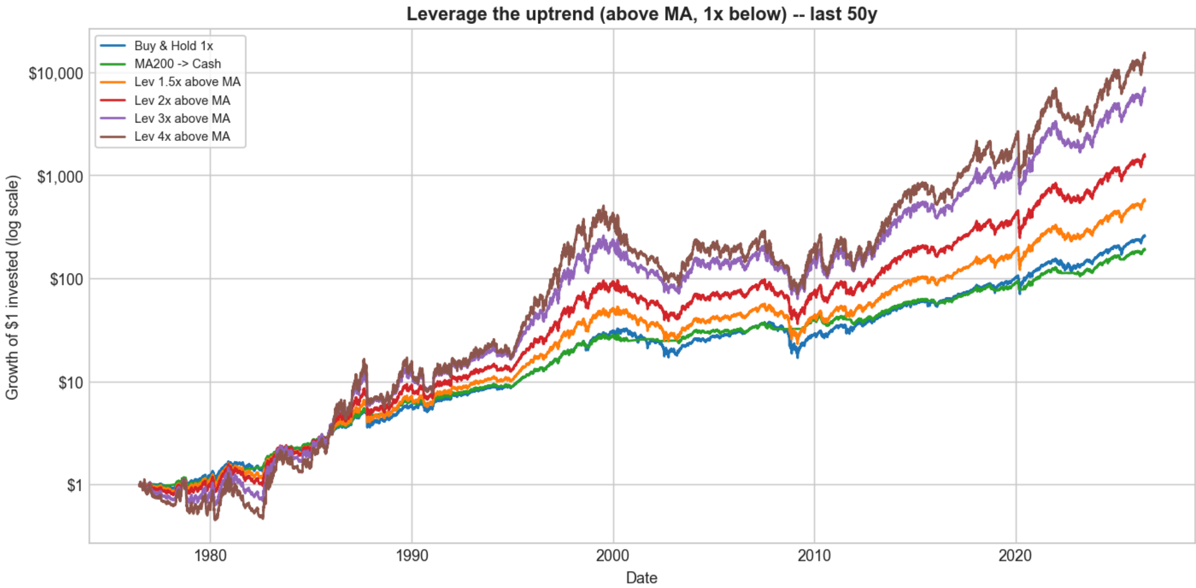
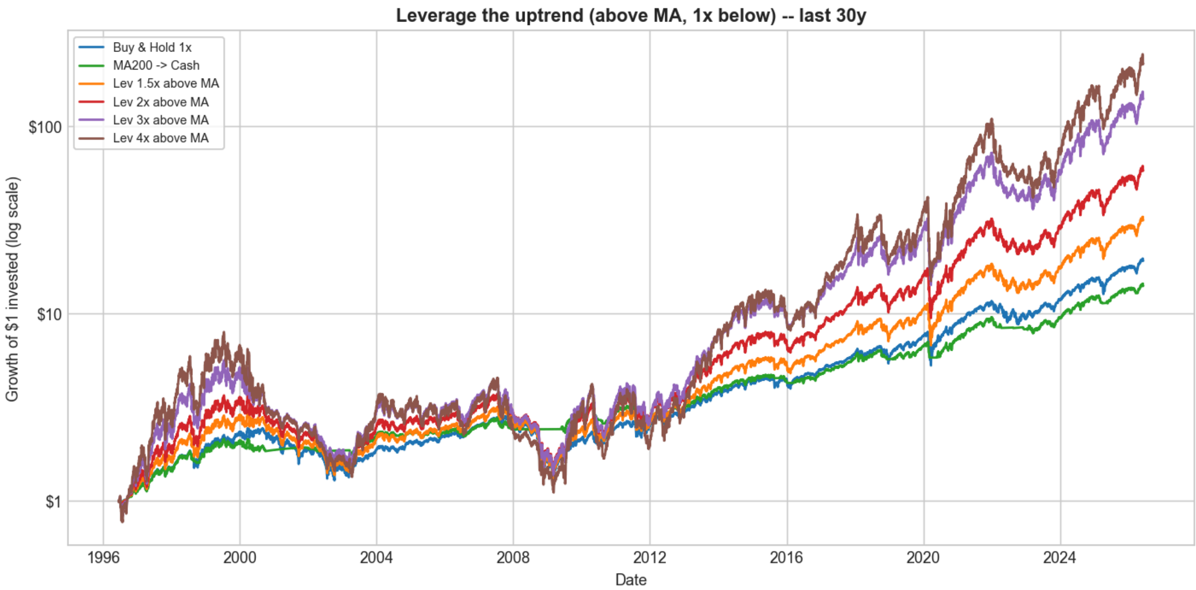
5. The switching strategy: leverage the uptrend

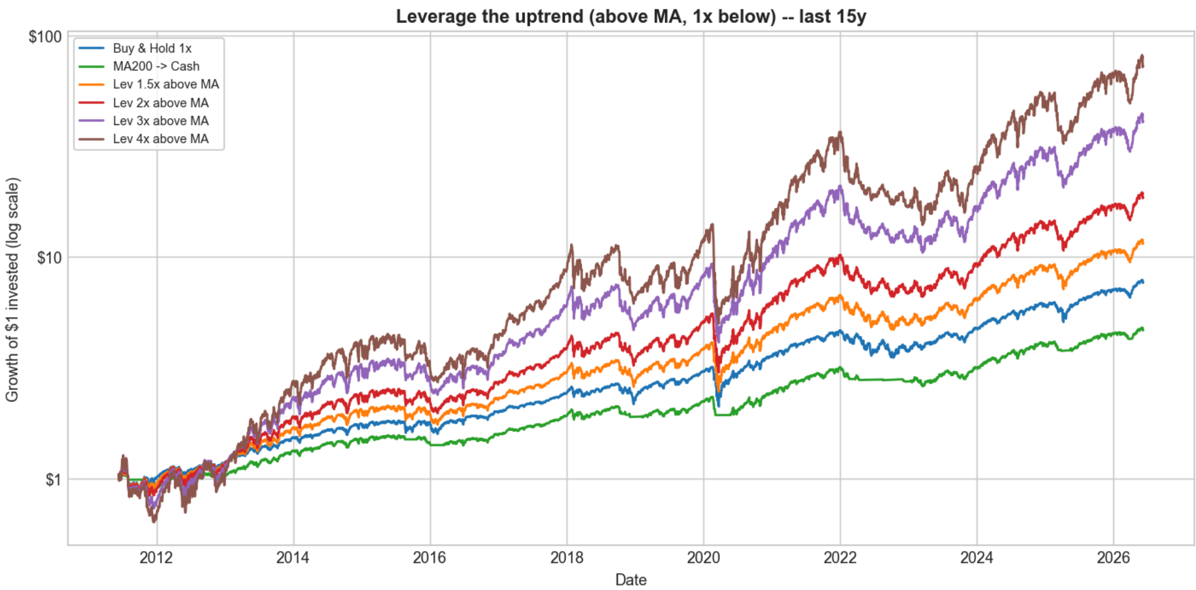
Use the 200-day MA to switch between leveraged and ordinary exposure: **hold L× leverage while the market is ABOVE the MA** — the calm, rising regime where the one-directional gains seen in §3 happen — and drop back to plain 1× while it is below. Each strategy is shown over the full history and the last 50 / 30 / 15 years (net of costs; the volatility column lets you reproduce the Sharpe):

Horizon	Strategy	Grew $1 to	CAGR	Vol	Sharpe	Sortino	Calmar	Max DD	IR vs S&P
full (1928+)	Buy & Hold 1x	$13,021	10.1%	18.9%	0.43	0.62	0.12	-83.9%	—
full (1928+)	MA200 -> Cash	$25,922	11.0%	12.6%	0.63	0.90	0.24	-46.4%	0.00
full (1928+)	Lev 1.5x above MA	$70,781	12.2%	23.6%	0.47	0.66	0.14	-85.7%	0.47
full (1928+)	Lev 2x above MA	$655,939	14.8%	28.9%	0.51	0.72	0.17	-89.0%	0.53
full (1928+)	Lev 3x above MA	$17,050,454	18.7%	40.4%	0.55	0.77	0.20	-95.5%	0.56
full (1928+)	Lev 4x above MA	$87,028,073	20.7%	52.5%	0.56	0.80	0.21	-98.8%	0.57
last 50y	Buy & Hold 1x	$256.7	11.7%	17.4%	0.49	0.69	0.21	-55.3%	—
last 50y	MA200 -> Cash	$189.4	11.1%	11.7%	0.60	0.85	0.54	-20.6%	-0.11
last 50y	Lev 1.5x above MA	$567.3	13.5%	21.8%	0.50	0.70	0.23	-59.0%	0.42
last 50y	Lev 2x above MA	$1,533	15.8%	26.7%	0.53	0.74	0.25	-62.8%	0.48
last 50y	Lev 3x above MA	$6,623	19.2%	37.4%	0.55	0.77	0.25	-75.8%	0.51
last 50y	Lev 4x above MA	$14,106	21.1%	48.6%	0.55	0.78	0.24	-86.1%	0.52
last 30y	Buy & Hold 1x	$19.1	10.4%	19.1%	0.50	0.70	0.19	-55.3%	—
last 30y	MA200 -> Cash	$14.0	9.2%	12.1%	0.61	0.84	0.45	-20.6%	-0.14
last 30y	Lev 1.5x above MA	$31.5	12.2%	23.4%	0.52	0.72	0.21	-59.0%	0.43
last 30y	Lev 2x above MA	$58.0	14.5%	28.4%	0.54	0.76	0.23	-62.8%	0.49
last 30y	Lev 3x above MA	$140.6	18.0%	39.2%	0.56	0.78	0.24	-75.8%	0.52
last 30y	Lev 4x above MA	$216.0	19.7%	50.6%	0.57	0.79	0.23	-86.1%	0.53
last 15y	Buy & Hold 1x	$7.68	14.6%	17.3%	0.79	1.11	0.43	-33.8%	—
last 15y	MA200 -> Cash	$4.68	10.9%	11.8%	0.81	1.11	0.61	-18.0%	-0.32
last 15y	Lev 1.5x above MA	$11.5	17.8%	21.7%	0.79	1.10	0.44	-40.1%	0.61
last 15y	Lev 2x above MA	$18.6	21.6%	26.7%	0.81	1.12	0.47	-45.8%	0.68
last 15y	Lev 3x above MA	$41.1	28.2%	37.5%	0.81	1.12	0.50	-56.0%	0.72
last 15y	Lev 4x above MA	$73.2	33.2%	48.7%	0.81	1.10	0.52	-64.5%	0.73







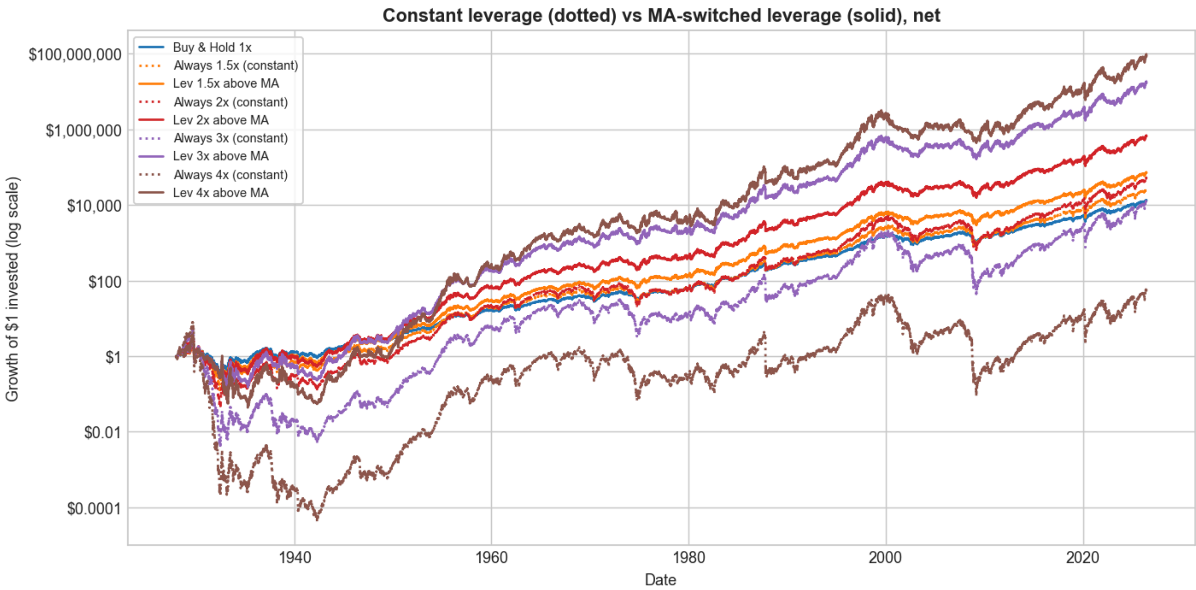


Reading the table. Leveraging the uptrend beats buy & hold on CAGR, Sortino and Calmar at every horizon, and matches its Sharpe in recent windows (last 15y: Sharpe ≈ 0.81 at 2–3×, same as buy & hold, but far higher CAGR). The **information ratio vs the S&P is strongly positive and rises with leverage** (0.47 → 0.57 over the full century, up to 0.73 over the last 15 years), so the leverage is adding genuine benchmark-relative return — note that plain MA→cash has a **negative** IR in recent windows (–0.11 to –0.32): it has *lagged* the index since the bull began. The cost is drawdown, which deepens with leverage (–86% at 1.5× to –99% at 4× over the full century).

6. Does the MA switch add value over just holding leverage?

Is the switching doing the work, or could you just hold constant leverage? Holding each level *constantly* (dotted) versus only **above the MA** (solid), 1928–2026:

Strategy	Grew $1 to	CAGR	Vol	Sharpe	Sortino	Max DD	IR vs S&P
Buy & Hold 1x	$13,021	10.1%	18.9%	0.43	0.62	-83.9%	—
Always 1.5x (constant)	$25,037	10.9%	28.4%	0.39	0.56	-94.9%	0.31
Lev 1.5x above MA	$70,781	12.2%	23.6%	0.47	0.66	-85.7%	0.47
Always 2x (constant)	$49,549	11.6%	37.8%	0.40	0.56	-98.5%	0.36
Lev 2x above MA	$655,939	14.8%	28.9%	0.51	0.72	-89.0%	0.53
Always 3x (constant)	$12,094	10.1%	56.8%	0.40	0.57	-99.9%	0.38
Lev 3x above MA	$17,050,454	18.7%	40.4%	0.55	0.77	-95.5%	0.56
Always 4x (constant)	$54	4.2%	75.7%	0.40	0.57	-100.0%	0.39
Lev 4x above MA	$87,028,073	20.7%	52.5%	0.56	0.80	-98.8%	0.57



The switch adds enormous value at every level: the MA-switched version has higher CAGR, Sharpe **and** a shallower drawdown than holding the same leverage all the time. Constant leverage runs out of road as it rises — constant 3× merely *matches* buy & hold ($12,094 vs $13,021) despite triple the volatility, and constant 4× **collapses to $54** (CAGR 4.2%, a –100% interim drawdown). The switched versions grow $1 to $656k (2×) and **$87 million** (4×). Constant leverage piles on volatility (Sharpe stuck near 0.40) for little extra compound return; the trend filter is what makes leverage pay.

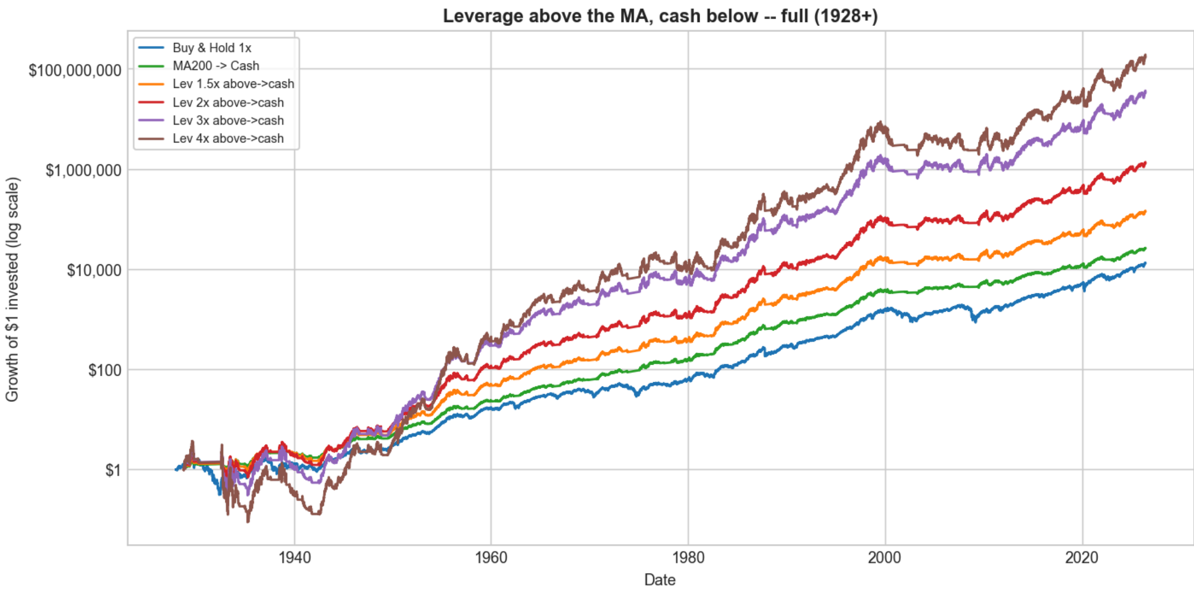
7. Leverage → cash: sidestep the downturns

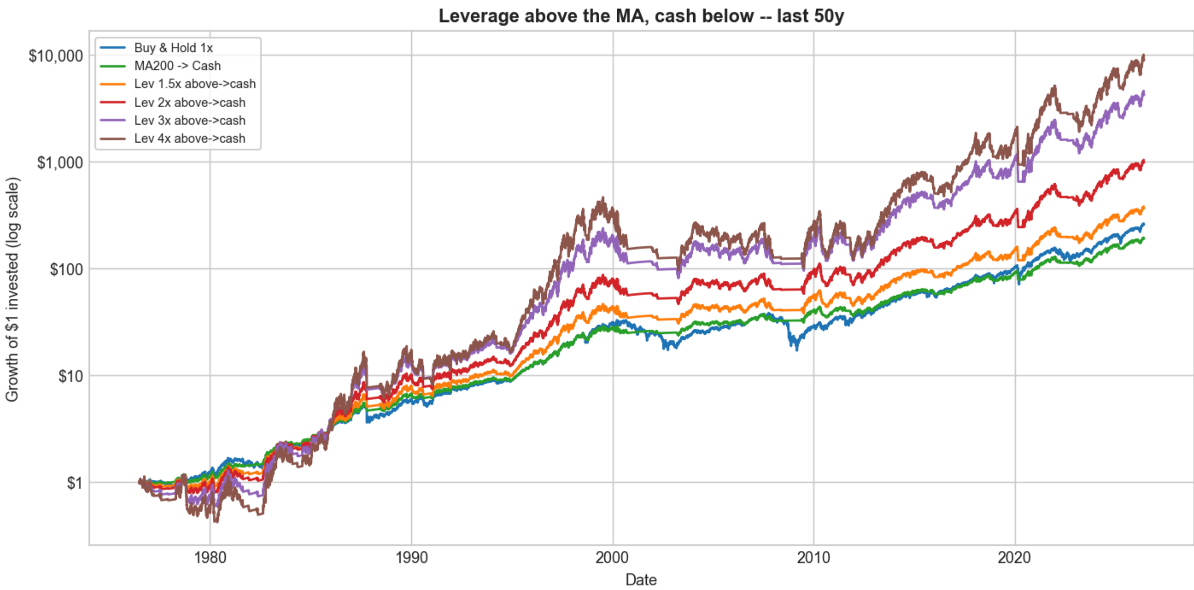
Instead of dropping to plain 1× below the MA, go all the way to **cash**. Over full / 50 / 30 / 15-year horizons (net):

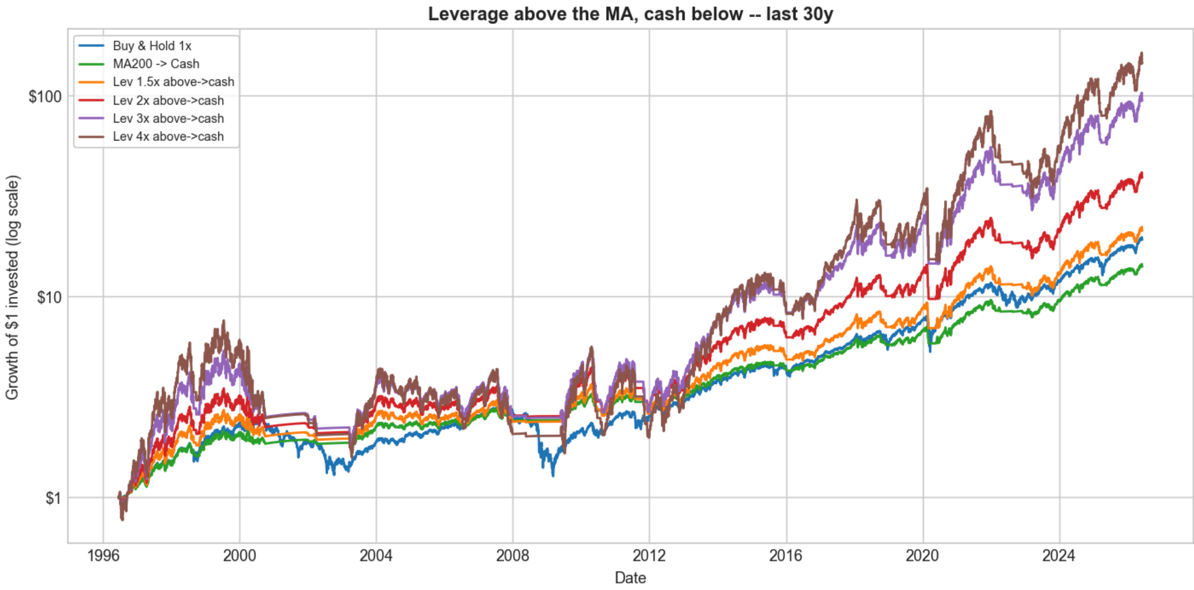
Horizon	Strategy	Grew $1 to	CAGR	Vol	Sharpe	Sortino	Calmar	Max DD	IR vs S&P
full (1928+)	Buy & Hold 1x	$13,021	10.1%	18.9%	0.43	0.62	0.12	-83.9%	—
full (1928+)	MA200 -> Cash	$25,922	11.0%	12.6%	0.63	0.90	0.24	-46.4%	0.00
full (1928+)	Lev 1.5x above->cash	$140,875	13.0%	18.9%	0.57	0.80	0.21	-62.9%	0.17
full (1928+)	Lev 2x above->cash	$1,305,593	15.6%	25.3%	0.57	0.81	0.21	-74.8%	0.34
full (1928+)	Lev 3x above->cash	$33,940,576	19.5%	37.9%	0.58	0.81	0.22	-89.8%	0.48
full (1928+)	Lev 4x above->cash	$173,246,846	21.5%	50.5%	0.58	0.82	0.22	-97.6%	0.52

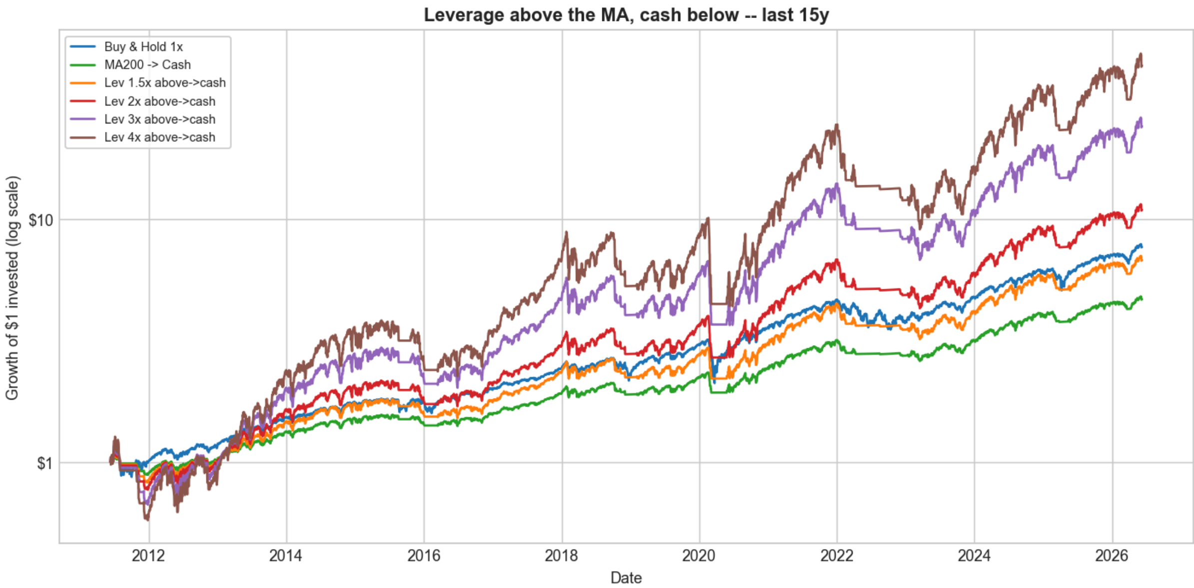
last 50y	Lev 2x above->cash	$988.6	14.8%	23.4%	0.53	0.74	0.32	-46.9%	0.23
last 50y	Lev 4x above->cash	$9,101	20.0%	46.9%	0.54	0.75	0.25	-79.3%	0.45
last 30y	Lev 2x above->cash	$39.1	13.0%	24.2%	0.54	0.74	0.28	-46.9%	0.18
last 30y	Lev 4x above->cash	$145.4	18.1%	48.4%	0.54	0.75	0.23	-79.3%	0.43
last 15y	Lev 2x above->cash	$10.9	17.3%	23.5%	0.73	1.00	0.47	-37.0%	0.21
last 15y	Lev 4x above->cash	$43.0	28.6%	47.0%	0.74	1.01	0.45	-63.0%	0.57

(Full per-horizon detail: `results/faber_lev_cash_horizons.csv`*.)*









Going to cash (not 1×) below the trend **shallows the drawdown** versus leverage-to-1× (2× above→cash: −75% over the full century, and only −37% / −47% over the last 15 / 30 years, against the leverage-to-1× −46% / −63%). The price is that in the recent bull its Sharpe (≈ 0.73 over 15 years) dips *below* buy & hold's (0.79), because the cash periods missed dips that quickly recovered.

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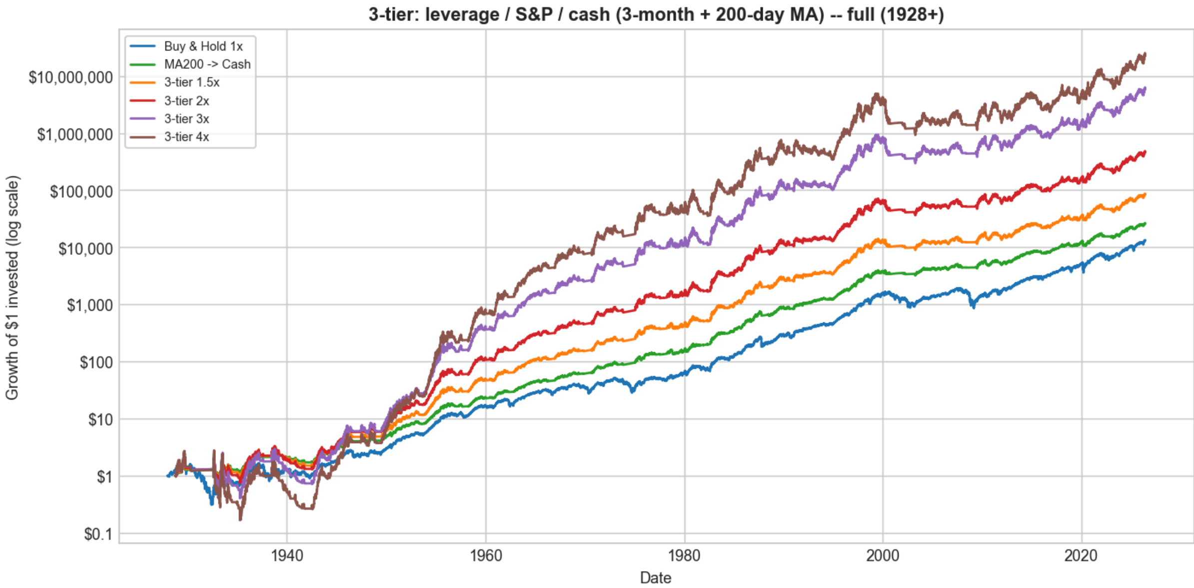
8. A three-tier rule: leverage → S&P → cash

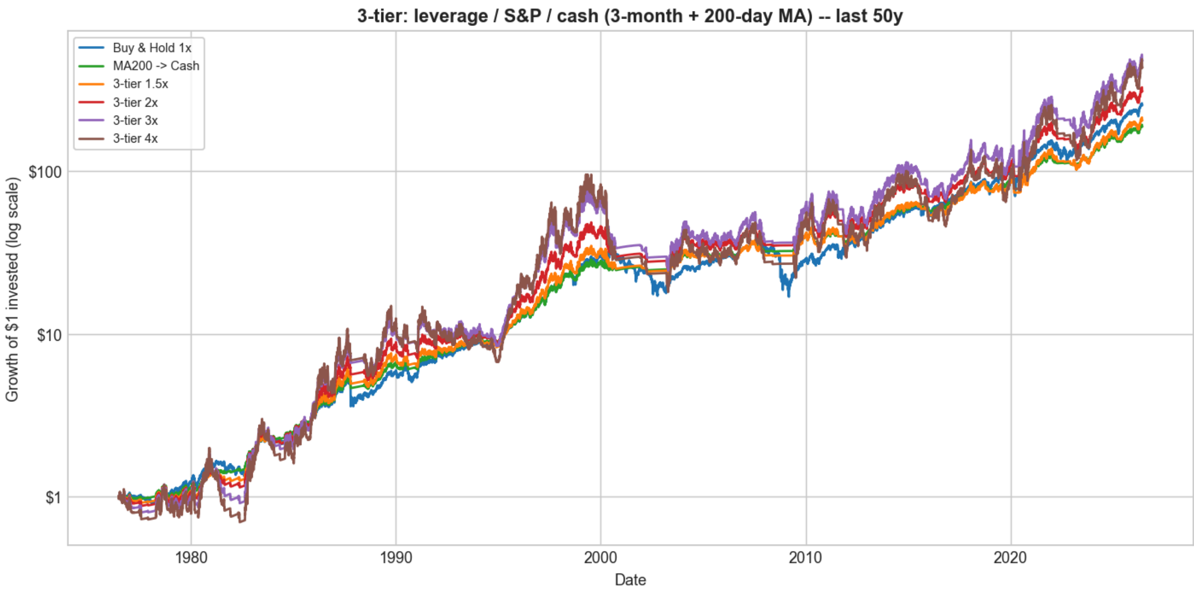
A finer version uses a fast (3-month) MA *and* the slow (200-day) MA: **leverage** above both, **plain 1× S&P** on a mild pullback (below the 3-month but above the 200-day), **cash** below the 200-day. Over full / 50 / 30 / 15-year horizons (net):

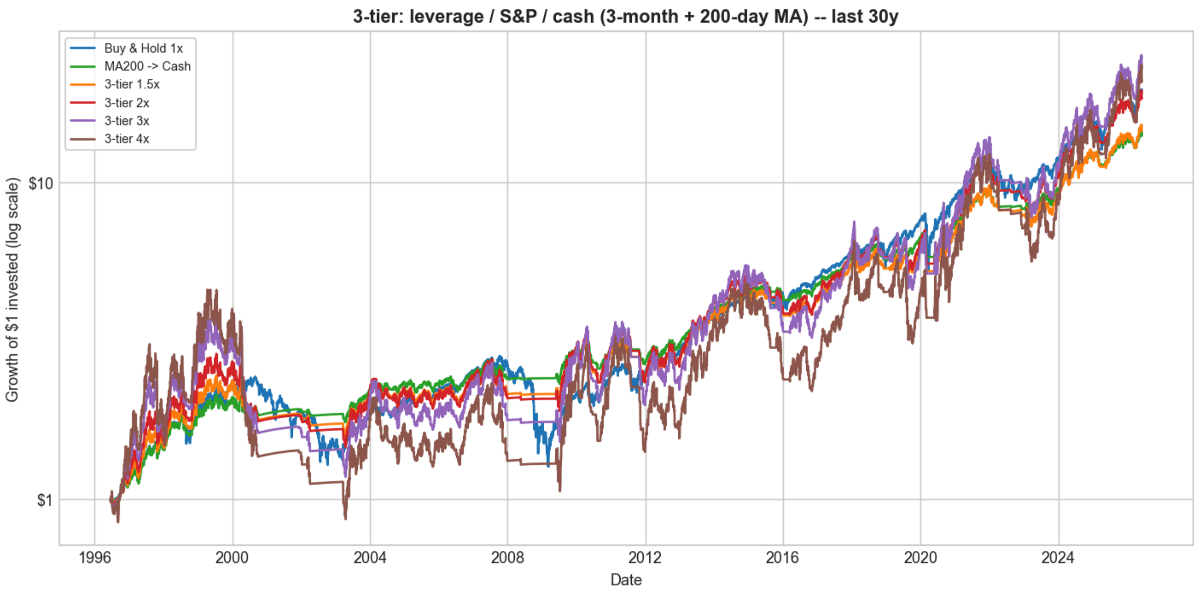
Horizon	Strategy	Grew $1 to	CAGR	Vol	Sharpe	Sortino	Calmar	Max DD	IR vs S&P
full (1928+)	MA200 -> Cash	$25,922	11.0%	12.6%	0.63	0.90	0.24	-46.4%	0.00
full (1928+)	3-tier 1.5x	$83,682	12.4%	17.4%	0.57	0.80	0.21	-58.3%	0.12
full (1928+)	3-tier 2x	$460,493	14.3%	22.5%	0.57	0.80	0.21	-67.8%	0.26
full (1928+)	3-tier 3x	$5,842,068	17.4%	33.0%	0.55	0.78	0.21	-84.1%	0.40
full (1928+)	3-tier 4x	$22,624,488	19.0%	43.6%	0.55	0.77	0.20	-94.0%	0.45
last 30y	3-tier 2x	$18.4	10.2%	21.4%	0.46	0.63	0.21	-49.3%	0.02
last 15y	Buy & Hold 1x	$7.68	14.6%	17.3%	0.79	1.11	0.43	-33.8%	—
last 15y	3-tier 2x	$5.95	12.7%	20.9%	0.60	0.82	0.39	-32.3%	-0.06

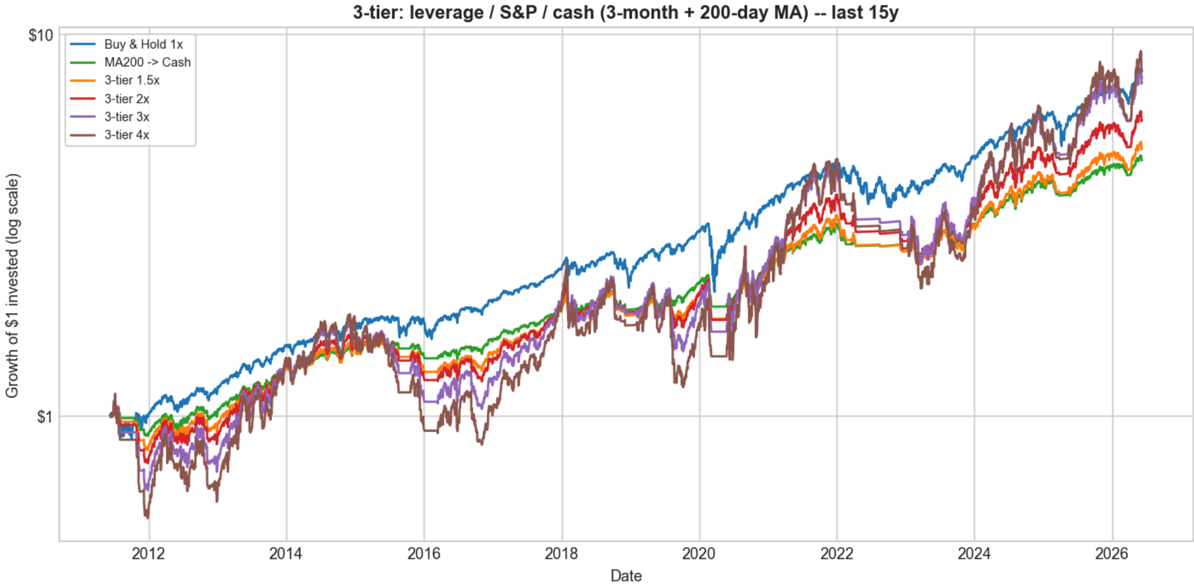
last 15y	3-tier 4x	$8.05	15.0%	40.4%	0.51	0.69	0.27	-54.5%	0.22

(Full per-horizon detail: `results/faber_lev_3tier_horizons.csv`*.)*









The 3-tier rule has the shallowest drawdowns of any leveraged variant over the full century (1.5×: –58%) and the best risk-adjusted profile *over the long run*. But the extra de-risking **hurts in the recent bull** — over the last 15 years its Sharpe falls to 0.51–0.65 (below buy & hold's 0.79) and even 4× barely beats 1×, because it sat in cash/1× through dips that recovered fast. It is the most conservative of the leveraged rules: best when bear markets are real and sustained, worst when every dip is bought.

An interesting cross-cutting result. The three leveraged rules rank differently by *era*. Over the full 1928+ history (which contains real, sustained bears), the more defensive rules (3-tier, leverage→cash) have the better risk-adjusted numbers. Over the **last 15 years** (a bull with only fast, V-shaped dips), the *least* defensive rule — straight leverage-above-MA — wins, because any time spent in cash or 1× was time out of a recovering market. There is no single best variant; the right amount of de-risking depends on whether bear markets persist.

Educational research only — not investment advice. Reproduce with `python run_faber_leverage.py`; figures are in `charts/`, numbers in `results/`.